Week 4: Transfer Learning & Image Segmentation

Stand on the Shoulders of Giants

IIT Madras

Live Coding Session — Week 4

Transfer Learning: Don't Start from Scratch

Take a model trained on millions of images, freeze its features, replace its head

Step 1: Load Pretrained

Conv Block 1 (edges)

Conv Block 2 (textures)

Conv Block 3...N (objects)

Trained on 1M+ ImageNet images

Classifier → 1000 classes
(ImageNet categories)

EfficientNet-B0: 5.3M params
Already knows edges, textures, shapes

→

Step 2: Freeze + Replace

Conv Block 1 ❄ FROZEN

Conv Block 2 ❄ FROZEN

Conv Block 3...N ❄ FROZEN

requires_grad = False

NEW Classifier → 3 classes
(pizza, steak, sushi)

Only train ~3,000 new params
Features stay, classifier adapts

→

Step 3: Train (fast!)

Same 5-step training loop
CrossEntropyLoss + Adam

~5 epochs
vs 50+ from scratch

High accuracy
even with small dataset

Leverage millions of images
you never had to label

Why it works: Early conv layers learn universal features (edges, textures) that transfer across tasks. Only the classifier needs to be task-specific.

New Task: Image Segmentation

Classification asks "what?" — Segmentation asks "what AND where?"

Classification (Weeks 1-3)

Image → "Sneaker"

1 image → 1 label

What is in the image?

Output: single class

→

Segmentation (Today!)

Image → Pixel Mask

1 image → 1 mask (H×W)

Where is it in the image?

Output: class per pixel

New Concepts Today

ConvTranspose2d

Upsampling — the reverse of pooling

Skip Connections

Encoder → Decoder detail recovery

BCEWithLogitsLoss

Per-pixel binary classification

(Same 5-step loop, different task!)

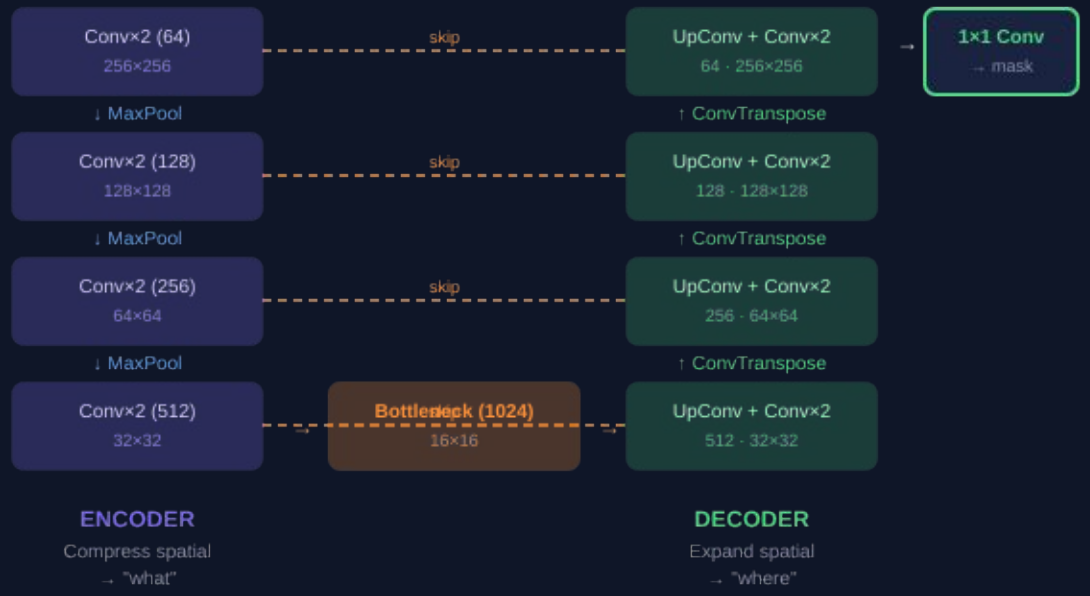
The challenge: CNNs compress spatial info (28×28 ⟶ 3×3). But segmentation needs to output at the original resolution.

The solution: U-Net — an encoder-decoder architecture that compresses then expands back, with skip connections to recover spatial detail.

Same building blocks: Conv2d, ReLU, MaxPool2d from week 3 — plus ConvTranspose2d for upsampling.

The U-Net Architecture

Encoder compresses, decoder expands, skip connections preserve detail



U-Net Key Ideas

Encoder

Conv blocks + MaxPool ↓

Same as week 3 CNN!

Bottleneck

Compressed representation

Knows "what" but not "where"

Decoder

ConvTranspose2d ↑ (upsample)

Recovers spatial resolution

Skip Connections

Encoder → Decoder (concatenate)

Preserves fine-grained details

that pooling would lose

`nn.ConvTranspose2d(in_ch, out_ch, kernel_size=2, stride=2)` — the reverse of MaxPool. Doubles H and W.

Today's Coding Session

~18 min

Transfer Learning

EfficientNet-B0 on food images. Freeze features, replace head, train 5 epochs.

~17 min

U-Net Segmentation

Build encoder-decoder with skip connections. Predict pixel masks on Oxford Pets.

+ 10 min Q&A at the end

Questions?

Week 4: Transfer Learning & Segmentation

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